Math 343 / 643 Fall 2026
Midterm Examination One

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March 4, 2026

Full Name _____

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Cheating Using or attempting to use unauthorized assistance, material, or study aids in examinations or other academic work or preventing, or attempting to prevent, another from using authorized assistance, material, or study aids. Example: using an unauthorized cheat sheet in a quiz or exam, altering a graded exam and resubmitting it for a better grade, etc.

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_____ signature

_____ date

Instructions

This exam is 75 minutes (variable time per question) and closed-book. You are allowed **one** page (front and back) of a "cheat sheet", blank scrap paper (provided by the proctor) and a graphing calculator (which is not your smartphone). Please read the questions carefully. Within each problem, I recommend considering the questions that are easy first and then circling back to evaluate the harder ones. No food is allowed, only drinks.

Problem 1 We are modeling the number of successful sales out of a known m daily cold calls made by a sales team. We observe the number of sales over n days, denoted by $\boldsymbol{x} := \{x_1, \ldots, x_n\}$, where each $x_i \in \{0, 1, \ldots, m\}$.

We suspect a "hurdle" effect: some days are complete busts due to market conditions (0 sales), but if the team gets past that hurdle and makes at least one sale, the total number of sales follows a Binomial distribution. We use a *Binomial Hurdle* model. With probability θ_1, a day yields strictly 0 sales. With probability $1 - \theta_1$, the day yields $x_i > 0$ sales, drawn from a Zero-Truncated Binomial (ZTBin) distribution,

$$X \sim \text{ZTBin}(m, \theta) := \frac{\binom{m}{x} \theta^x (1-\theta)^{m-x}}{1 - (1-\theta)^m} \mathbb{1}_{x \in \{1,2,\ldots,m\}}, \quad \mathbb{E}[X] = \frac{m\theta}{1 - (1-\theta)^m}$$

Thus, the DGP for the observed $X_1, \ldots, X_n$ is iid:

$$\begin{aligned} \mathbb{P}(X = 0) &= \theta_1 \\ \mathbb{P}(X = y) &= (1 - \theta_1)\, \text{ZTBin}(m, \theta_2) \end{aligned}$$

Let $n_0 :=$ be the number of days with exactly 0 sales, and $n_+ :=$ be the number of days with strictly > 0 sales (such that $n_0 + n_+ = n$). Also let $s := \sum_{\{i: x_i > 0\}} x_i$.

(a) [2 pt / 2 pts] What are the parameter(s) in this model?

θ_1, θ_2

(b) [6 pt / 8 pts] Write the likelihood function f for all n observations.

$$\begin{aligned} f(\boldsymbol{x} \mid \theta_1, \theta_2) &= \prod_{\{i: x_i = 0\}} \theta_1 \prod_{\{i: x_i > 0\}} (1 - \theta_1) \frac{\binom{m}{x_i} \theta_2^{x_i} (1 - \theta_2)^{m - x_i}}{1 - (1 - \theta_2)^m} \\ &= \theta_1^{n_0} \left(\frac{1 - \theta_1}{1 - (1 - \theta_2)^m} \right)^{n_+} \left(\prod_{\{i: x_i > 0\}} \binom{m}{x_i} \right) \left(\frac{\theta_2}{1 - \theta_2} \right)^s (1 - \theta_2)^{m n_+} \end{aligned}$$

(c) [3 pt / 11 pts] Assume Laplace priors for the parameter(s). Write those priors below. If they are improper, then write them as proportionalities below.

$$\theta_1 \sim \mathrm{U}\,(0,\,1)\,, \quad \theta_2 \sim \mathrm{U}\,(0,\,1)$$

(d) [5 pt / 16 pts] Derive the kernel of the conditional posterior distribution of θ_1 given θ_2 and the data. Use the priors from (c). Don't forget the indicator function for θ_1.

$$f(\theta_1 \mid \theta_2, \boldsymbol{x}) \propto \theta_1^{n_0}\,(1-\theta_1)^{n_+}\,\mathbb{1}_{\theta_1 \in (0,1)}$$

(e) [2 pt / 18 pts] Is the kernel in the previous question correspond to a known distribution? If so, provide that distribution below. If not, write "no" below.

$$f(\theta_1 \mid \theta_2, \boldsymbol{x}) = \mathrm{Beta}\,(n_0+1,\, n_+ + 1)$$

(f) [6 pt / 24 pts] Derive the kernel of the conditional posterior distribution of θ_2 given θ_1 and the data. Use the priors from (c). Don't forget the indicator function for θ_2.

$$\begin{aligned} f(\theta_2 \mid \theta_1, \boldsymbol{x}) &\propto \left(\frac{1}{1-(1-\theta_2)^m}\right)^{n_+} \left(\frac{\theta_2}{1-\theta_2}\right)^{s} (1-\theta_2)^{mn_+}\,\mathbb{1}_{\theta_2 \in (0,1)} \\ &= (1-(1-\theta_2)^m)^{-n_+}\,\theta_2^{s}\,(1-\theta_2)^{mn_+ - s}\,\mathbb{1}_{\theta_2 \in (0,1)} \end{aligned}$$

(g) [2 pt / 26 pts] Is the kernel in the previous question correspond to a known distribution? If so, provide that distribution below. If not, write "no" below.

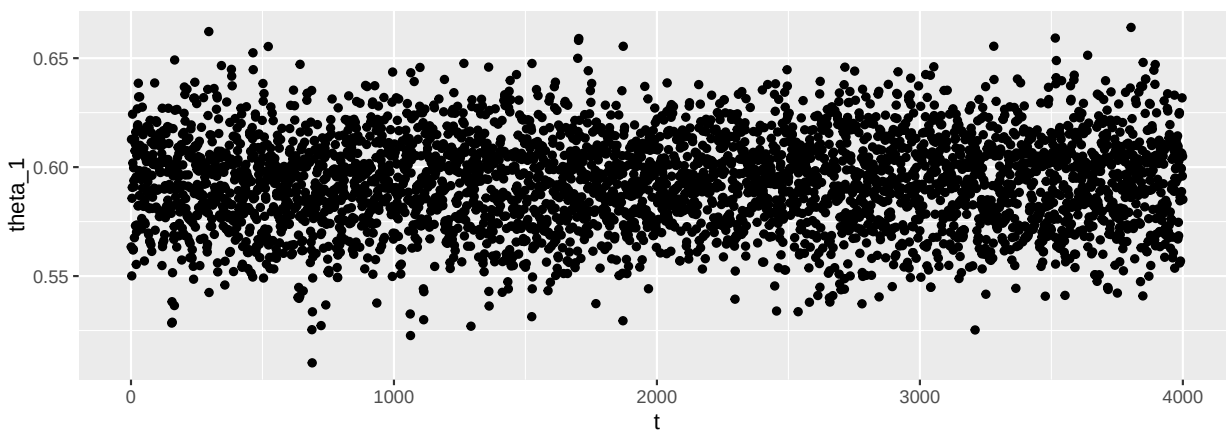
No

(h) [3 pt / 29 pts] As of now, can we use a Gibbs Sampler to draw samples from the posterior distribution? Yes/no and why?

No, because we need to be able to sample from all conditional distributions. We can only sample from the first.

Regardless of what you wrote for answers in (e) and (g), we implement a sampler for the posterior distribution and we sample 2,000 draws. In our example, $m = 15$.

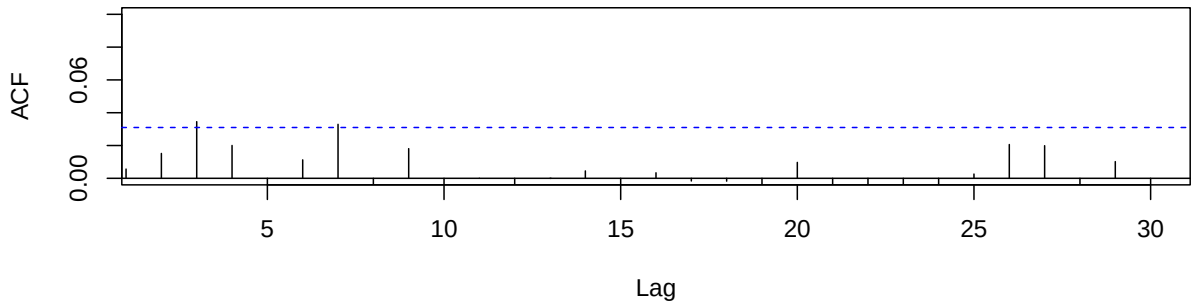
(i) [3 pt / 32 pts] Below is a scatterplot where the x denotes sampling iteration number the y axis displays samples of θ_1 given $\boldsymbol{x}$.



Do we need to burn a certain number of initial samples? If yes, write the approximate iteration number. If no, write zero.

0

Below is an autocorrelation plot for the samples of θ_1 given $\boldsymbol{x}$.

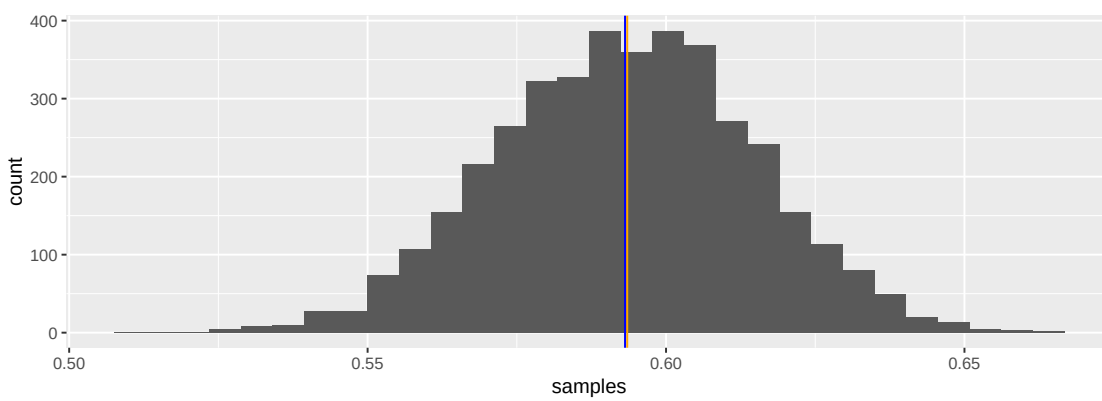


(j) [3 pt / 35 pts] Do we need to thin a certain number of initial samples? If yes, write the approximate iteration number. If no, write zero.

8

Regardless of what you wrote for answers in (h) and (i), assume we do the necessary burning and thinning.

Below are the samples of θ_1 given $\boldsymbol{x}$.



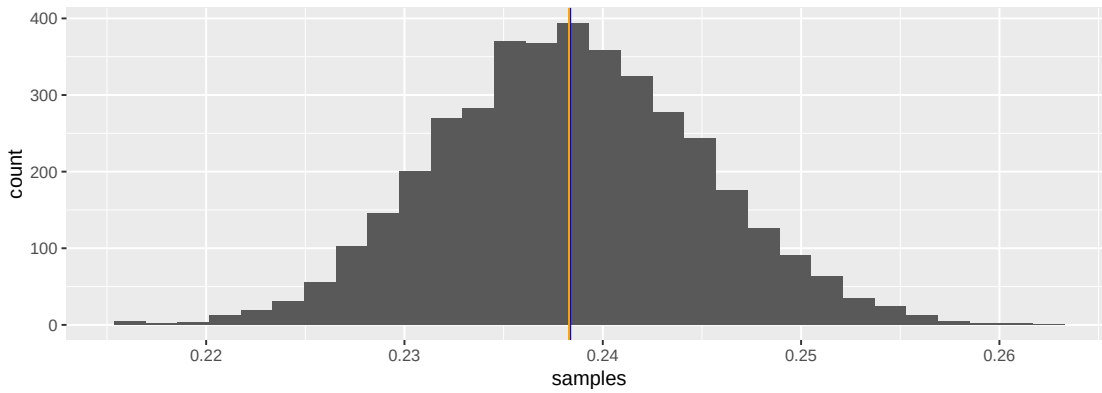
(k) [3 pt / 38 pts] Provide a $\text{CR}_{\theta_1,95\%}$.

$[.550, .635]$

(l) [4 pt / 42 pts] Decision $H_0 : \theta_1 \leq 0.5$ at $\alpha = 5\%$ and provide justification.

Reject H_0 because the Bayesian p-value is $\mathbb{P}(H_0 \mid \boldsymbol{x}) = \mathbb{P}(\theta_1 \leq 0.5 \mid \boldsymbol{x}) \approx 0 < \alpha$.

Below are the samples of θ_2 given $\boldsymbol{x}$.



(m) [5 pt / 47 pts] Decision $H_0 : \theta_2 = 0.5$ at $\alpha = 5\%$ and provide justification.

Reject H_0 because the Bayesian p-value is $\mathbb{P}(H_0 \mid \boldsymbol{x}) = \mathbb{P}(\theta_1 \in [0.5 \pm \delta] \mid \boldsymbol{x}) \approx 0 < \alpha$ for any reasonable δ.

(n) [5 pt / 52 pts] Estimate $\mathbb{E}[X]$ to the nearest two decimals.

$$\begin{aligned}
\mathbb{E}[X] &= \mathbb{E}[X \mid X = 0]\,\mathbb{P}(X = 0) + \mathbb{E}[X \mid X > 0]\,\mathbb{P}(X > 0) \\
&\approx (0)\hat{\hat{\theta}}_1^{\text{MMSE}} + \left(\frac{m\hat{\hat{\theta}}_2^{\text{MMSE}}}{1 - (1 - \hat{\hat{\theta}}_2^{\text{MMSE}})^{15}}\right)(1 - \hat{\hat{\theta}}_1^{\text{MMSE}}) \\
&= \left(\frac{15 \cdot .238}{1 - (1 - .238)^{15}}\right)(1 - .59) = 1.49
\end{aligned}$$

(o) [5 pt / 57 pts] Extra credit: Explain what you would do to test $H_0 : \theta_1 \leq \theta_2$ at level $\alpha = 5\%$.

Given the sampling chain where the first column is the samples of θ_1 and the second column is θ_2, I would create a third column with values equal to $\theta_1 - \theta_2$. I would then create a histogram of that column and estimate the proportion of samples less than zero. If the proportion is less than 5%, I would reject H_0 (otherwise retain H_0).

Problem 2 Consider the following observations where the + sign indicates censoring. The unit of measurements here is seconds.

```
0.012 0.030+ 0.039+ 0.064 0.073+ 0.153 0.203+ 0.349 0.493+ 0.844
```

(a) [5 pt / 62 pts] First, ignore censoring. Create a table with the values of the t_i's, d_i's and n_i's. Label the n_i values as n_0, n_1, n_2, etc. If the value of the entry is zero or not applicable, you can leave the cell in the table blank. The first row is filled in for you.

t_i	d_i	n_i
0	0	$n_1 = 10$
0.012	1	$n_2 = n_1 - 1 = 9$
0.030	1	$n_3 = n_2 - 1 = 8$
0.030	1	$n_4 = n_3 - 1 = 7$
0.064	1	$n_5 = n_4 - 1 = 6$
0.073	1	$n_6 = n_5 - 1 = 5$
0.153	1	$n_7 = n_6 - 1 = 4$
0.203	1	$n_8 = n_7 - 1 = 3$
0.349	1	$n_9 = n_8 - 1 = 2$
0.493	1	$n_{10} = n_9 - 1 = 1$
0.844	1	$n_{11} = n_{10} - 1 = 0$

(b) [5 pt / 67 pts] Create a 95% CI for survival past 0.5 seconds rounded to the nearest millisecond.

$$
\begin{aligned}
CI_{S(0.5),95\%} &= \left[\hat{\hat{S}}(0.5) \pm 1.96\sqrt{\frac{\hat{\hat{S}}(0.5)(1-\hat{\hat{S}}(0.5))}{n}} \right] \\
&= \left[0.1 \pm 1.96\sqrt{\frac{0.1(1-0.9)}{10}} \right] \\
&= [-0.086, 0.286]
\end{aligned}
$$

(c) [5 pt / 72 pts] [i] What is wrong with the interval from (b)? [ii] Why did this happen? [iii] Could you use the bootstrap to fix this problem? [iv] What new problems would the bootstrap introduce?

i The lower bound is below zero. Survival times must be non-negative.

ii Low sample size. $n = 10$ which is too small for the CLT to kick in.

iii Yes

iv The bootstrap is also approximate relying on large sample asymptotics. We have low sample size. So the bootstrap interval won't have accurate coverage.

(d) [5 pt / 77 pts] Create a table with the values of the t_i's, s_i's, d_i's, q_i's and n_i's. Label the n_i values as n_0, n_1, n_2, etc. If the value of the entry is zero or not applicable, you can leave the cell in the table blank. The first row is filled in for you.

t_i	s_i	d_i	q_i	n_i
0		0		$n_1 = 10$
0.012		1		$n_2 = n_1 - 1 - 1 - 1 = 7$
	0.030		1	
	0.039		1	
0.064		1		$n_3 = n_2 - 1 - 1 = 5$
	0.073		1	
0.153		1		$n_4 = n_3 - 1 - 1 = 3$
	0.203		1	
0.349		1		$n_5 = n_4 - 1 - 1 = 1$
	0.493		1	
0.844		1		$n_6 = n_5 - 1 = 0$

(e) [8 pt / 85 pts] Graph the Kaplan-Meier survival curve estimate on the gridlines below. Mark censored observations with a vertical dash on the survival curve estimate. Use open dots and closed dots where appropriate. *Label the axes appropriately.* If you do not know who to do this, draw as much as you know how to do so you can answer the following questions in this problem.

To generate the Kaplan-Meier survival curve estimate, we need to compute the step function's points of discontinuity which are the values of the product limit estimator for all t_i:

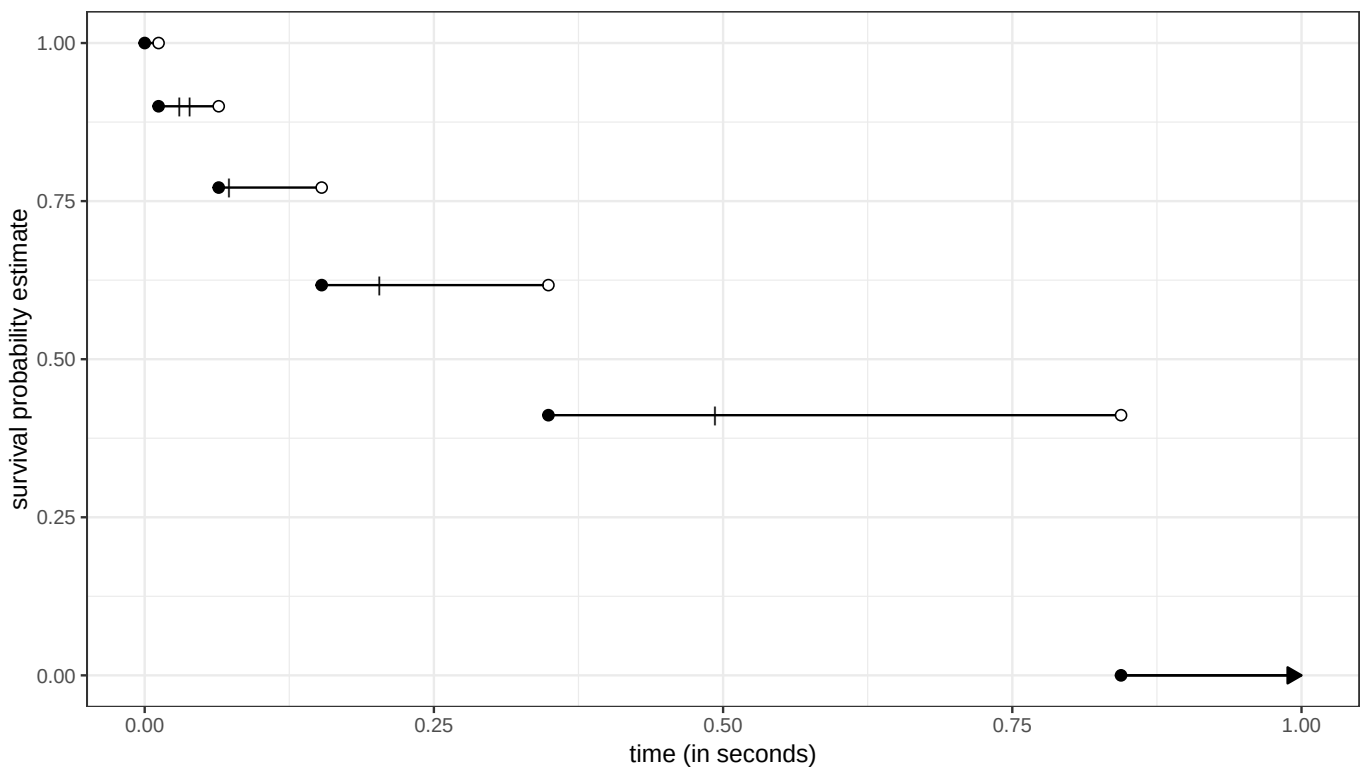
$$\hat{S}(t_1) = \hat{S}(0.012) = p_1 = 1 - \frac{d_1}{n_1} = 1 - \frac{1}{10} = 0.9$$

$$\hat{\hat{S}}(t_2) = \hat{\hat{S}}(0.064) = p_1 p_2 = p_1 \left(1 - \frac{d_2}{n_2}\right) = 0.9 \left(1 - \frac{1}{7}\right) = 0.9 \cdot 0.857 = 0.771$$

$$\hat{\hat{S}}(t_3) = \hat{\hat{S}}(0.153) = p_1 p_2 p_3 = p_1 p_2 \left(1 - \frac{d_3}{n_3}\right) = 0.771 \left(1 - \frac{1}{5}\right) = 0.771 \cdot 0.8 = 0.617$$

$$\hat{\hat{S}}(t_4) = \hat{\hat{S}}(0.349) = p_1 p_2 p_3 p_4 = p_1 p_2 p_3 \left(1 - \frac{d_4}{n_4}\right) = 0.617 \left(1 - \frac{1}{3}\right) = 0.617 \cdot 0.667 = 0.411$$

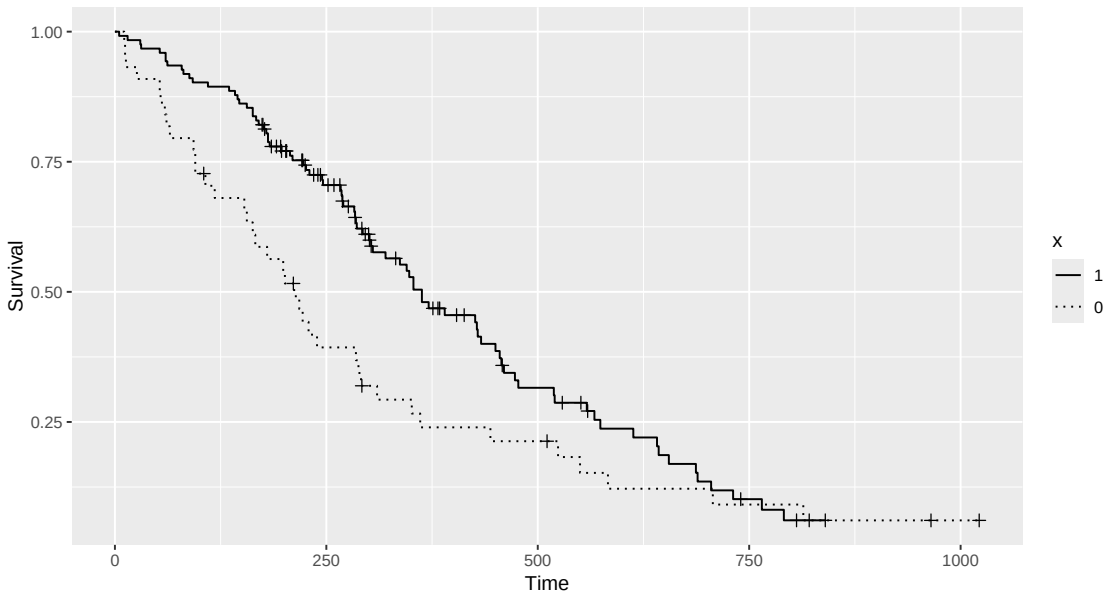
$$\hat{\hat{S}}(t_5) = \hat{\hat{S}}(0.844) = p_1 p_2 p_3 p_4 p_5 = p_1 p_2 p_3 p_4 \left(1 - \frac{d_5}{n_5}\right) = 0.411 \left(1 - \frac{1}{1}\right) = 0.411 \cdot 0 = 0$$



(f) [2 pt / 87 pts] Using the plot from (e), estimate the median survival. If you didn't put an answer in (e), make up a legal survival curve and answer based on that.

0.349, the y corresponding to the minimum time $\hat{\hat{S}}(y) < 0.5$.

Problem 3 We are interested in the differential survival (measured in days) of two groups of lung cancer patients differentiated by a cutoff of their beginning Karnovsky performance score rated by a physician. Group 1 is defined as those above the cutoff which have "good performance" and group 0 is defined as those below the cutoff which have "bad performance". There are $n_1 = 123$ subjects with good performance and $n_0 = 44$ subjects with bad performance. Below are the KM survival curve estimates for both groups. The vertical dash signs indicate a censored observation. In the legend, "x" means "group". Let Y_0 denote the SDGP for group 0 and let Y_1 denote the SDGP for group 1.



(a) [2 pt / 89 pts] Estimate $S_{Y_1}(500d) - S_{Y_0}(500d)$

10%

(b) [2 pt / 91 pts] Using just a visual inspection of the KM survival curve estimates, make a decision about $H_0 : \text{Med}[Y_0] = \text{Med}[Y_1]$. Justify your answer.

Reject H_0 because the median survival in group 1 looks significantly larger than the median survival in group 0.

Below is the output from the function `survdiff`:

```
          N Observed Expected (O-E)^2/E (O-E)^2/V
x=0  44      38     27.2      4.29      5.65
x=1 123      82     92.8      1.26      5.65
```

(c) [3 pt / 94 pts] Make a principled decision about $H_0 : f_{Y_0}(y) = f_{Y_1}(y)$. Justify your answer fully.

Using the log rank test at $\alpha = 5\%$, $\hat{\hat{\phi}} = 4.29 + 1.26 = 5.7 > 1.96^2 = 3.84 \Rightarrow$ reject H_0.

Problem 4 Fisher introduced the permutation test (often called the randomization test) in his 1935 book, The Design of Experiments. The dataset he famously used was Charles Darwin's data on the heights of Zea mays (corn) plants. Darwin had grown 15 pairs of corn plants: one cross-fertilized and one self-fertilized. He grew them in the same pots to control for environmental differences. Below is the original data Fisher used:

i	Group 1: Cross-fertilized (inches)	Group 2: Self-fertilized (inches)
1	23.500	17.375
2	12.000	20.375
3	21.000	20.000
4	22.000	20.000
5	19.125	18.375
6	21.500	18.625
7	22.125	18.625
8	20.375	15.250
9	18.250	16.500
10	21.625	18.000
11	23.250	16.250
12	21.000	18.000
13	22.125	12.750
14	23.000	15.500
15	12.000	18.000
Average	20.192	17.575

We will ignore the pairing of the data (since we didn't discuss this in class) so we will not be duplicating Fisher's original statistical test. We will be doing a permutation test and assuming a test statistic of $\bar{x}_1 - \bar{x}_2$.

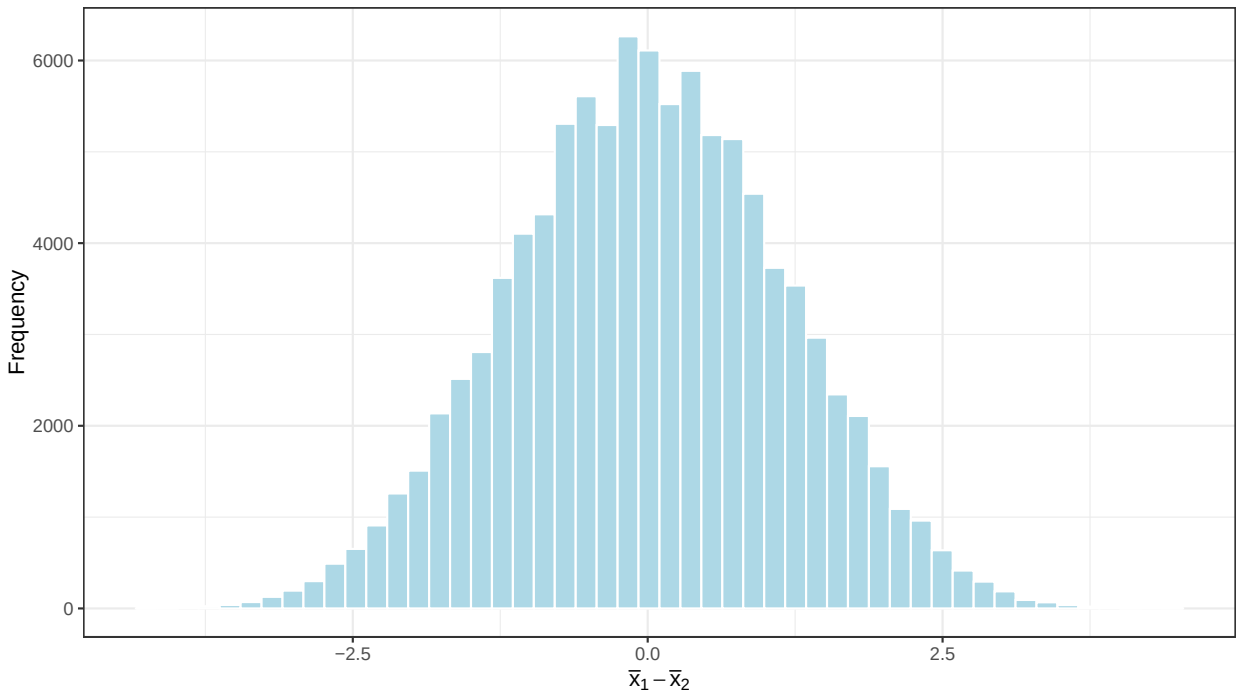
(a) [2 pt / 96 pts] How many permutations exist in this scenario?

$\binom{30}{15}$

(b) [3 pt / 99 pts] What would be a potential weakness of using the test statistic $\bar{x}_1 - \bar{x}_2$?

If the DGP's for group 1 and group 2 have equal expectations but different variances, this test would have very little power.

Below is a histogram of 100,000 test statistics each computed on different permutations of the data.



(c) [3 pt / 102 pts] At $\alpha = 5\%$, test $H_0 : \text{DGP}_1 = \text{DGP}_2$. Justify your answer fully.

The 95% retainment region seems to be at $[-2.5, 2.5]$ and the observed statistic is 20.192 - 17.575 = 2.617 hence we reject H_0.

(d) [3 pt / 105 pts] Estimate Fisher's p-value for the test in (c). Your answer must comport with (c) for full credit.

Since 2.617 is just barely greater than the edge of the retainment region, integrating the area above would be slightly less than 2.5% maybe $\approx$ 2%. Multiplying by two for the two sided test would give us an estimate of 4%.